Constant Current LED Power Supply

SDS60-1400IL

Product description

SDS60-1400IL is an indoor DALI constant current LED driver that can provide 700/750/800/850/900/950/1000/1050/1100/1150/1200/1250/1300/1350/1400mA 15 levels of current , the output current can be easily set through DIP Switch, and complies with the DALI IEC62386. Use the DALI main control/Touch DIM (Push DIM) switch to dim the light, with smooth curves and no flicker. Its input voltage range is 220-240Vac, with a conversion efficiency of up to 88%. It works in the natural cooling case temperature range of -20°C~+45°C, and has ultra-high power factor and ultra-low total harmonics. Distortion, low standby power consumption, and all-round protection functions not only greatly improve the reliability of the product, but also ensure the product life cycle.

Standards

EN61347-1:2015
EN 61347-2-13:2014+A1
AS/NZS61347.2.13
EN 61347-2-13:2014 +A1
EN61347-1:2015
IEC62386

Characteristics

- 15 levels of current
- With active PFC function
- IP20
- Suitable for LED indoor lighting, such as panel lights
- 1-100% full dimming without visible flicker, 4K high frequency PWM dimming exempt from assessment level
- Comply with DALI IEC62386

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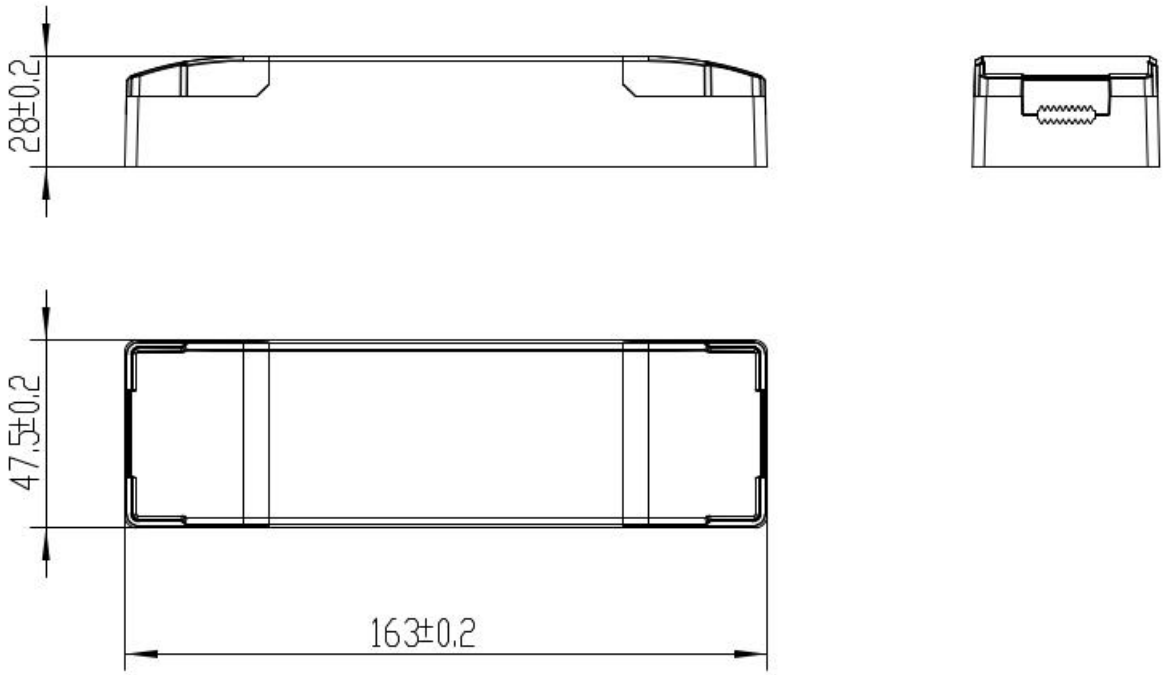
Specifications

Model		SDS60-1400IL
Output	turn on time(S)	＜0.7
	output power(W)	60
	output voltage(V)	20-50
	output voltage tolerance	±8%
	ripple voltage	5%
	Line Regulation	±5%
	Load Regulation	±5%
	working current range(A)	0.7-1.4
	SVM	≤0.4
	Pst	≤1
	dimming type	DALI2.0
	dimming range	1-100%
Input	rated DC supply voltage(Vdc)	-
	rated supply voltage(Vac)	220-240
	voltage range(Vac)	198-264
	line frequency(Hz)	50/60
	input current(A)	0.32
	efficiency	≥88%@full load
	average efficiency 3 3	≥87%
	no load power consumption(W)	≤0.5W
	power factor	0.95@full load
	Displacement factor	0.95
	THD(typ.) THD（）	10
	inrush current(Ipk)（Ipk）	46A
	Leakage current	5mA
	short circuit protection	hiccup mode, restart automatically after fault correction.
	over load protection	exceed maximum rated load times 1.1

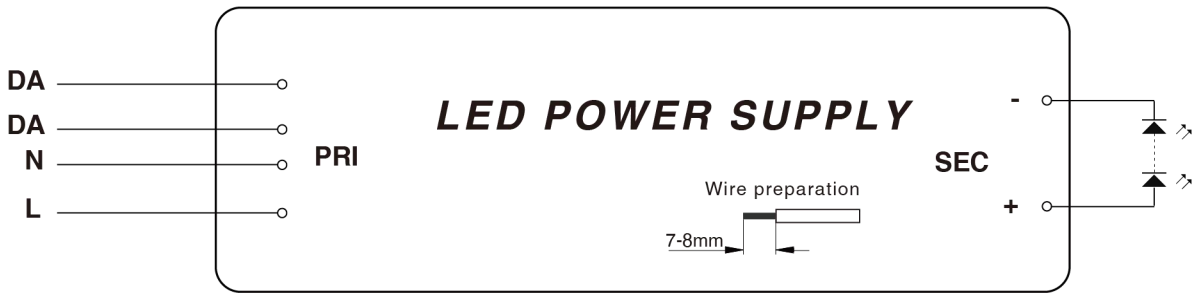
Protection	Over temperature protection	-
	surge capacity	L-N: 1KV
	Withstand voltage	Input-Output：3750V/5mA/1min
Ambient and Life	Ta(C)	-20…45
	Tc max.(C)	max.85
	Storage Temperature(C)	-40...85
	ambient humidity range	10%...90%RH, Not condensing
	nominal life-time(hrs)	50'000@Ta
Other	dimensions (L×W×H)(mm)	163*47.5*28
	weight(g)	275g
	casing material	Plastic
	housing colour	White
	type of protection	IP20
	protection class	class II
	certificate	
Note	1.Tolerance:includes set up tolerance, line regulation and load regulation. 2.Tested at full load,230Vac.Refer to"Power Factor" and "EFFICIENT"curve graphs. 3.Calculate the model's average efficiency for each test voltage by testing at 100%, 75%, 50%, and 25% of rated current and then computing the simple arithmetic average of these four values. 4.All parameters NOT specially mentioned are measured at nominal voltage input, rated load and 25 of ambient temperature. 5.The power supply is considered as a component that will be operated in combination with final equipment. Since EMC performance will be affected by the complete installation, the final equipment manufacturers must re-qualify EMC Directive on the complete installation again.	

Dimensions (mm)

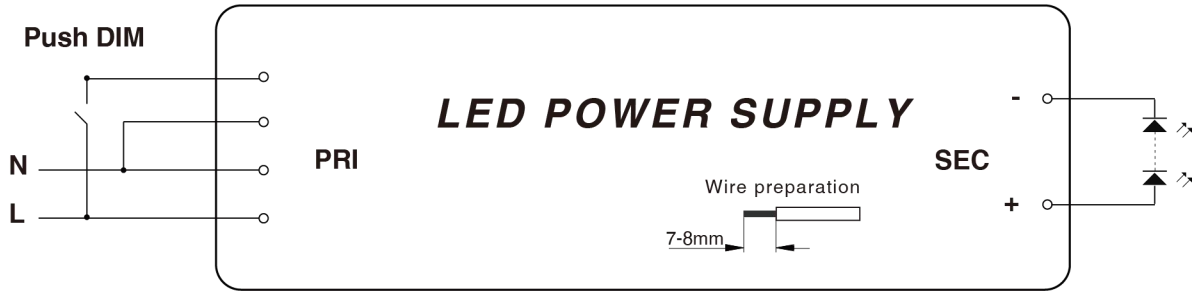


Wiring Diagram

DALI



Push DIM



AC	GY350,4P, 2*0.75—1.5mm2
DALI	GY350,4P, 2*1.5mm2
DC	GY350 2P 2*0.75mm2

supply to the DALI interface through a reset switch;

Short press: Press the reset button quickly and then let go. The button will automatically return to the disconnected state. The duration of pressing the button is 40~500ms;

Long press: Press the reset button quickly without letting go, the duration is >500ms;

Serie	Function	Operation	Description
1	Memory	Power on	In Push mode, after the LED driver is powered on, it will remember the state before shutting down; If it was in standby state before shutting down, it will still be in standby state after power on. After short pressing the switch next time, the default will be maximum brightness.
2	ON/OFF	Short press	If the light is on, short press to turn off the light; if the light is off, short press to turn on the light, and the brightness will be the same as before the last time you turned off the light.
3	Dimming	Long press	When the reset switch is long pressed, it enters the dimming state and starts to change after long pressing for 500ms; If it was brightened the previous time, it will be dimmed next time. After releasing the reset button, the dimming will stop and the current brightness will be maintained; The fade time from 1% to 100% is 3 seconds; If you press and hold in the light-off state, the brightness will gradually increase from the minimum brightness;
4	Synchronization	Long press >10s	When the long press time reaches 10 seconds, it will jump to 100% output for multi-light synchronization.

DIP switch settings

PIN				Iout	Vout
1	**2**	**3**	**4**	**mA**	**V**
ON	OFF	OFF	OFF	700	35-50
OFF	**ON**	OFF	OFF	750	35-50
ON	**ON**	OFF	OFF	800	30-50
OFF	OFF	**ON**	OFF	850	30-50
ON	OFF	**ON**	OFF	900	30-50
OFF	**ON**	**ON**	OFF	950	25-50
ON	**ON**	**ON**	OFF	1000	25-50
OFF	OFF	OFF	**ON**	1050	20-50
ON	OFF	OFF	**ON**	1100	20-50
OFF	**ON**	OFF	**ON**	1150	20-50
ON	**ON**	OFF	**ON**	1200	20-50
OFF	OFF	**ON**	**ON**	1250	20-46
ON	OFF	**ON**	**ON**	1300	20-44
OFF	**ON**	**ON**	**ON**	1350	20-40
ON	**ON**	**ON**	**ON**	1400	20-38

Electrical curves

Fig. 1 Output load-Temperature curve
1 -

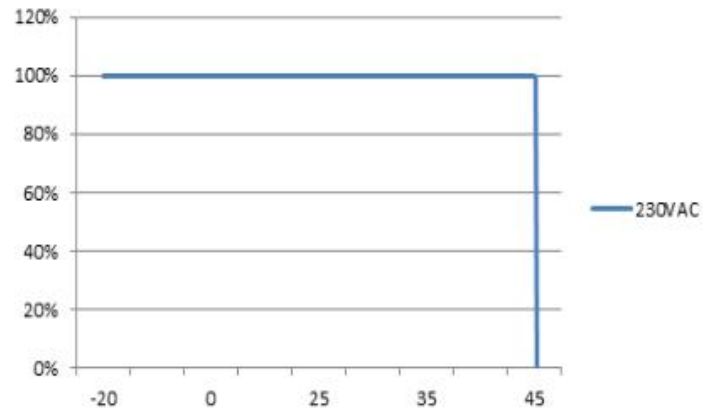


Fig. 2 Static characteristic curve
2

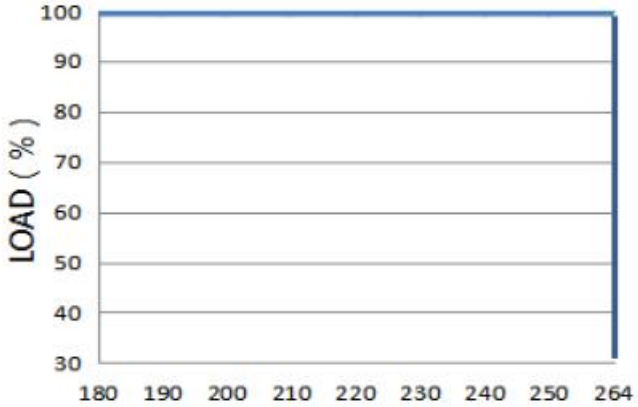


Fig. 3 curve

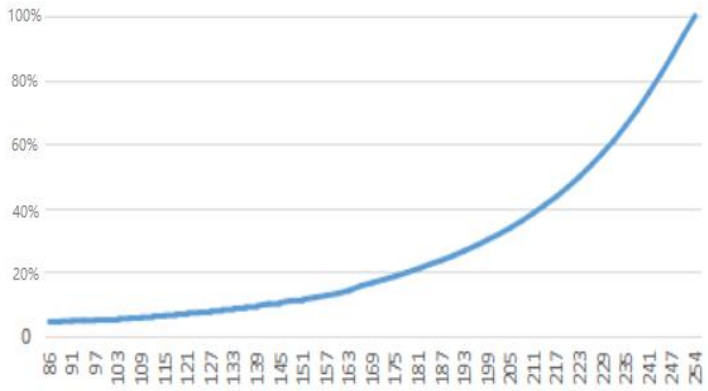


Fig. 4 Power factor characteristic curve
4

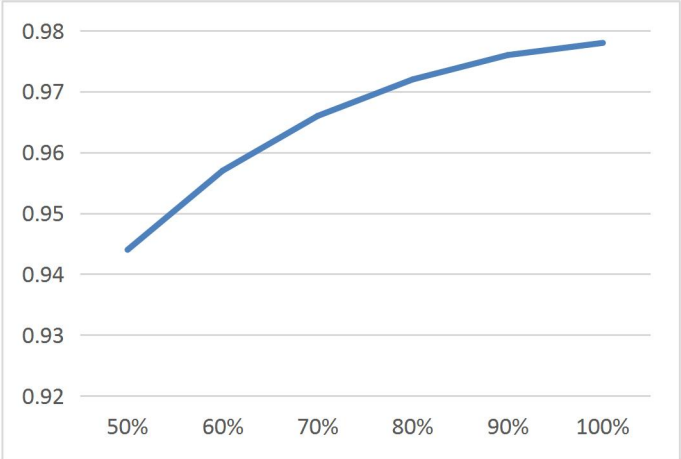


Fig.5 Total harmonic distortion curve（THD）
5 （THD）

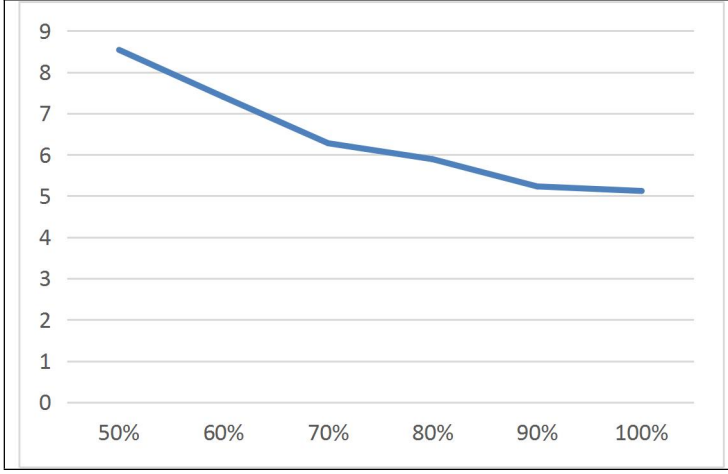
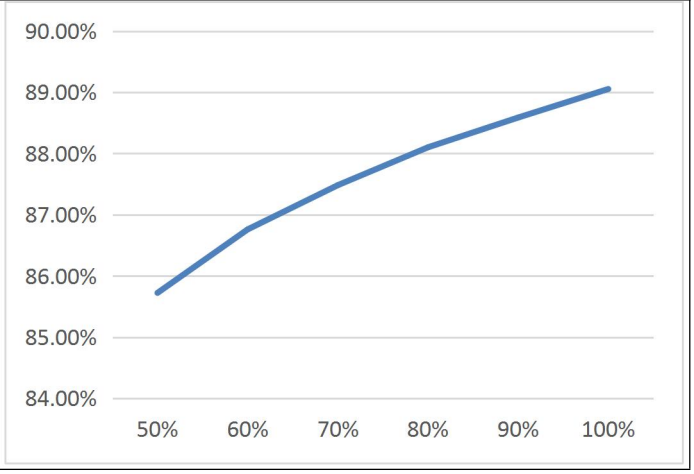


Fig.6 Efficiency-Load curve
6 -



MCBS

MCBS / Model	B10	B13	B16	B20	C10	C13	C16	C20
SDS60-1400IL	11	15	18	23	19	25	31	38

Package

Model	Carton quantity(pcs)	Carton dimension(mm)	G.W./CTN(kg)
SDS60-1400IL			

Revision history

Date	Rev.	Remark
2023.6.25	V0.01	Initial release.
2023.8.25	V1	Officially released